Contactless Palm Biometric Verification Architecture: Engineering Implementation Under Zero-Data and Permissive Open-Source Constraints

Legal Permissibility and Commercial Operational Viability of MediaPipe

The integration of computer vision pipelines into commercial security systems requires strict adherence to open-source software licensing and intellectual property protections. Google MediaPipe is distributed under the Apache License 2.0, a permissive software framework that grants broad, royalty-free permissions for commercial exploitation, continuous integration into proprietary codebases, binary modification, and commercial SaaS distribution[1]. Organizations may build and monetize enterprise authentication systems incorporating MediaPipe without payment thresholds or licensing fees[2]. Compliance requires maintaining standard legal attributions: retaining original copyright notices, enclosing a full text of the Apache 2.0 license, supplying applicable NOTICE documentation, and explicitly documenting modifications made to the source modules[2]. MediaPipe also provides an explicit patent grant from contributors, protecting commercial entities against patent infringement litigation concerning its core pipeline designs[2].





The operational mechanics of the MediaPipe Hands framework rely on a two-stage cascaded deep learning architecture designed for low-latency inference on edge devices[5]. The initial phase employs a single-shot detector known as BlazePalm, which is optimized to regress palm bounding boxes across the complete visual scene rather than locating whole articulated hands[6]. By localizing the rigid palm center, the system remains invariant to non-rigid finger flexion, wrist articulation, and hand rotations[6]. The secondary phase processes the cropped sub-image through a convolutional keypoint regression network that computes twenty-one 3D landmarks (

x, y, z coordinates) representing the physical coordinates of the skeletal joints, knuckles, and wrist boundary[6].
MediaPipe was originally engineered for hand-tracking, augmented reality interfaces, and sign language interpretation rather than biometric identity verification[6]. The twenty-one landmarks provide coarse geometric joint positions but do not capture the high-frequency epidermal ridge patterns, flexion creases, and principal lines of the palmar fascia required for high-entropy biometric identification[6]. Consequently, MediaPipe cannot function as an isolated biometric matching engine[6]. Its value lies in serving as an open-source front-end framework for real-time hand acquisition, pose gating, scale normalization, and dynamic Region of Interest (ROI) extraction, handling the localization and alignment challenges typical of touchless computer vision[5].

Dynamic Palm ROI Extraction and Canonical Alignment

Traditional contactless palmprint recognition systems frequently encounter performance degradation during initial ROI segmentation[5]. Earlier image-processing approaches depended on morphological binarization, adaptive skin-color thresholding, and outer contour tracking to detect the inter-digital valley points between the fingers[5]. In touchless configurations without physical guide pegs, these heuristic edge algorithms routinely fail due to variable illumination, hand rotation, rings, fingernail polish, and motion blur[5].
Using MediaPipe's deep keypoint regressor addresses these vulnerabilities by establishing a stable, keypoint-anchored coordinate system directly over the palmar surface[5].

Joint Landmark Index	Anatomical Joint Identifier	Geometric Function within ROI Pipeline
0	Carpal Wrist Center (P_0)	Baseline anchor defining vertical hand axis and proximal boundary[8].
1	Thumb Carpometacarpal (P_1)	Lateral ulnar/radial reference for palmar basin boundaries[8].
5	Index Metacarpophalangeal (P_5)	Radial boundary anchor defining upper palm line and baseline scale[8].
9	Middle	Central axis reference defining rotation angle and

	Metacarpophalangeal (P_9)	vertical crop center[8].
13	Ring Metacarpophalangeal (P_{13})	Distal anchor used for palmar arch curvature and planarity calculations[8].
17	Little Metacarpophalangeal (P_{17})	Ulnar boundary anchor for transverse scale and rotation normalization[8].

Dynamic palm ROI extraction follows a multi-stage geometric sequence:
To counteract hand roll and orientation skew across arbitrary camera perspectives, an alignment angle θ is calculated relative to the image coordinate frame[5]. The primary central axis vector v_{axis} connects the carpal wrist landmark $P_0 = (x_0, y_0)$ to the middle finger knuckle landmark $P_9 = (x_9, y_9)$, establishing the vertical orientation:

$$v_{\text{axis}} = P_9 - P_0 = (x_9 - x_0,\ y_9 - y_0)$$

$$\theta = \arctan 2\,(x_9 - x_0,\ -(y_9 - y_0))$$

Alternatively, the transverse vector connecting the index knuckle P_5 to the pinky knuckle P_{17} establishes the horizontal plane:

$$v_{\text{trans}} = P_{17} - P_5 = (x_{17} - x_5,\ y_{17} - y_5)$$

Using the computed angle θ , the entire image frame I_{raw} is rotated around P_9 via an affine transformation matrix, which aligns the middle finger parallel to the vertical axis and removes rotational variations[5].
Because camera-to-hand distances vary without physical restraints, spatial scale normalization is required to ensure scale invariance[5]. The transverse palm width L_{palm} is derived from the Euclidean distance between the aligned index knuckle P'_5 and pinky knuckle P'_{17} [9]:

$$L_{\text{palm}} = \|P'_{17} - P'_5\|_2 = \sqrt{(x'_{17} - x'_5)^2 + (y'_{17} - y'_5)^2}$$

The center coordinates of the biometric ROI, $C_{\text{roi}} = (x_c, y_c)$, are positioned within the

central sub-palmar hollow to capture the principal lines and secondary creases while avoiding the finger roots and carpal wrist bones[8]. The coordinates are defined as:

$$x_c = \frac{x'_5 + x'_{17}}{2}$$

$$y_c = y'_9 + \beta \cdot L_{\text{palm}}$$

Empirical research identifies the offset coefficient $\beta \in [0.45, 0.60]$ as optimal for positioning the crop window over the center of the palm[9]. A square sub-window with dimensions $S_{\text{roi}} \times S_{\text{roi}}$ is cropped around C_{roi}, where $S_{\text{roi}} = \gamma \cdot L_{\text{palm}}$ (with $\gamma \in [0.80, 1.00]$)[12]. The cropped sub-image is resampled to a uniform resolution (128×128 or 224×224 pixels) and equalized using Contrast-Limited Adaptive Histogram Equalization (CLAHE) to mitigate uneven lighting[7].

Pipeline Dimension	Classical Contour / Valley Detection	MediaPipe Landmark-Derived Dynamic ROI
Tracking Mechanics	Skin thresholding, convex hull scanning, radial distance local minima[5].	BlazePalm anchor proposals combined with 21-point skeletal regression[5].
Environmental Robustness	Fails under varying lighting, cluttered backgrounds, and low skin contrast[5].	Robust across skin tones, ambient illumination, and cluttered visual fields[5].
Tolerance to Occlusions	Compromised by rings, bracelets, bandage tape, and closed fingers[5].	Maintains knuckle landmark positions despite jewelry or finger posture changes[5].
Computational Overhead	High CPU consumption driven by iterative edge and contour tracing[5].	Low latency (≈ 3–8 ms execution on edge processors)[5].

Biometric Verification Strategies Under Zero-Image

Constraints

A core challenge in deploying deep learning biometrics without an initial operational database is the cold-start data acquisition problem. Training high-capacity convolutional models from scratch requires tens of thousands of annotated biometric samples to prevent overfitting[14]. To circumvent this limitation, systems can deploy unsupervised deterministic coding methods or utilize publicly available metric learning embeddings[12].

Unsupervised Deterministic Feature Extraction via Competitive Code

Competitive Code (CompCode) is an effective hand-crafted baseline for palmprint matching that operates without training datasets[13]. Drawing on neurophysiological evidence that cortical simple cells decompose visual inputs through orientation-selective receptive fields, CompCode filters line features by convolving the palm ROI with an oriented bank of 2D Gabor kernels[17]. Because palm creases represent negative valleys against the skin, the negative real component of the 2D Gabor filter is used[19]:

$$\psi(x, y, \theta_p) = -\frac{\omega}{\sqrt{2\pi}\kappa} \exp\left(-\frac{\omega^2}{8\kappa^2}(4x'^2 + y'^2)\right)\left(\cos(\omega x') - \exp\left(-\frac{\kappa^2}{2}\right)\right)$$

In this formulation, $x' = x\cos\theta_p + y\sin\theta_p$, $y' = -x\sin\theta_p + y\cos\theta_p$, ω represents the radial frequency bandwidth, and κ represents a constant derived from the half-amplitude filter response δ via $\kappa = \sqrt{2\ln 2}\frac{2^\delta+1}{2^\delta-1}$ [19]. The filter bank samples six orientations spaced across the semi-circular domain:

$$\theta_p = \frac{p\pi}{6}, \quad p \in \{0, 1, 2, 3, 4, 5\}$$

For each pixel coordinate (x, y) , an orientation competition determines the kernel that maximizes the valley response:

$$J(x, y) = \arg\min_p \{I_{\text{roi}}(x, y) * \psi(x, y, \theta_p)\}$$

The optimal orientation index $J(x, y)$ is encoded into a 3-bit binary Gray code, producing three binary feature bit-planes: $B_1(x, y)$, $B_2(x, y)$, and $B_3(x, y)$ [17]. Bit-level storage compresses a 128×128 palmprint template into approximately six kilobytes, facilitating rapid matching on low-power hardware[13].

Comparing a stored gallery template T against an input query template Q uses a normalized bitwise XOR Hamming distance[17]:

$$D_{\text{Hamming}}(T,Q) = \frac{1}{3NM}\sum_{k=1}^{3}\sum_{x=1}^{N}\sum_{y=1}^{M}\left(T_k(x,y) \oplus Q_k(x,y)\right)$$

To correct for residual positional errors introduced during touchless hand capture, the query template is translated across a local window $\Delta x, \Delta y \in \{-d, \ldots, d\}$ with $d \in [2,4]$ pixels[13]:

$$D_{\text{final}}(T,Q) = \min_{\Delta x, \Delta y} D_{\text{Hamming}}\left(T, Q^{(\Delta x, \Delta y)}\right)$$

Competitive Code requires no iterative machine learning, contains no learned weights that could trigger training-data license disputes, and yields genuine recognition rates exceeding 98.5% on benchmark datasets[13].

Deep Metric Feature Extraction via Pre-Trained Ensembles

The primary alternative to deterministic filtering employs deep convolutional neural networks trained via angular margin metric learning[12]. Lightweight architectures such as MobileFaceNet, ResNet-18, or CompNet (a hybrid architecture embedding learnable Gabor banks into initial convolutional layers) can be pre-trained using Additive Angular Margin Loss (ArcFace)[12]. ArcFace projects normalized feature vectors onto an open-set hypersphere, enforcing an angular margin m between identities to expand inter-class variance while penalizing intra-class spread[12]:

$$\mathcal{L}_{\text{ArcFace}} = -\log \frac{\exp\left(s \cdot \cos(\theta_{y_i} + m)\right)}{\exp\left(s \cdot \cos(\theta_{y_i} + m)\right) + \sum_{j \neq y_i} \exp\left(s \cdot \cos\theta_j\right)}$$

where s denotes the feature radius scale and θ_{y_i} represents the angle separating the feature embedding from the target identity weight vector[12].

Once pre-trained on open biometric corpora, the classification head is discarded[12]. The remaining convolutional backbone functions as a static feature extractor that maps an aligned 224×224 input ROI into a 512-dimensional unit vector f where $\|f\|_2 = 1$ [12]. Verification matches evaluate the Cosine Similarity between enrolled template f_T and query template f_Q [12]:

$$S(f_T, f_Q) = f_T \cdot f_Q$$

A verification match is confirmed if $S(f_T, f_Q) > \tau_{\text{thresh}}$, with the threshold configured to

meet institutional false accept and false reject criteria[12].

Hand Geometry Versus Epidermal Palmprint Texture

Biometric system design must balance the trade-offs between hand-geometry features (derived directly from landmark spatial configurations) and epidermal palmprint textures[7].

Hand geometry metrics—such as finger length ratios, knuckle aspect spans, and perimeter areas—possess limited biological entropy[23]. Hand geometry parameters show high phenotypic correlation across demographics, and single monocular RGB cameras introduce perspective distortions when fingers flex or tilt[5]. These limitations typically restrict hand-geometry verification systems to Equal Error Rates between 1.0% and 6.0%, which is inadequate for high-security access control[23].

In contrast, the detailed texture of the palmar surface—comprising principal lines, accessory creases, and dermal patterns—provides substantial biometric entropy, yielding Equal Error Rates below 0.1% under controlled evaluation[22]. Hand geometry derived from MediaPipe should therefore be restricted to pose validation, distance gating, and spoof detection, while identity verification matches rely on the underlying palmar texture[7].

Modality and Backbone	Discriminative Entropy	Typical Equal Error Rate (EER)	Training Requirement	Inference/ Match Latency	Sensitivity to Hand Pose
Hand Geometry (MediaPipe Landmarks)	Low ($\approx 10^2$ distinct states)[23].	$1.0\% - 6.0$ [23]	None (direct Euclidean coordinate metrics)[23].	< 1 ms	High (susceptible to finger flexion and tilt)[5].
Competitive Code (2D Gabor Bank)	High ($\approx 10^6$ distinct line variations)[17].	$0.05\% - 0.$ [13]	**Zero** (deterministic mathematical filtering)[16].	$\approx 10 - 15$ ms (CPU execution)[13]	Moderate (compensated by landmark alignment)[5].
CompNet (Learnable Gabor CNN)	Very High (directional optimized kernels)[21].	$0.04\% - 0.$ [21]	Pre-trained on public research benchmarks[21].	$\approx 15 - 25$ ms (CPU/NPU) [21]	Low (learned convolutional invariance)[21].

MobileFaceNet + ArcFace Loss	Very High (hyperspherical margin separation)[1][2].	$0.03\% - 0.$[22]	Pre-trained on public research benchmarks[12].	$\approx 12 - 20$ ms (CPU/NPU)[12]	Very Low (high resilience to translation)[1][2].

Dataset Lineage, Open Repositories, and Licensing Governance

Using publicly available computer vision models requires careful evaluation of training dataset provenance and software license boundaries[3].

Prominent research benchmarks include the Tongji Contactless Palmprint Dataset, which contains 12,000 images representing 600 palms across 300 subjects, captured in two distinct sessions separated by an average of 61 days[13]. The IIT Delhi (IITD) Touchless Palmprint Database contains 2,600 images from 235 participants captured in an unconstrained indoor environment[5]. The Hong Kong Polytechnic University (PolyU) Contactless 2D/3D Database provides paired multi-spectral and photometric samples[17].

These academic benchmarks are typically governed by restrictive End-User License Agreements (EULAs) limiting use to non-commercial scientific research[7]. Deploying neural weights trained directly on Tongji or PolyU data inside a commercial binary creates legal and intellectual property liabilities[7]. Commercial alternatives, such as the AxonData Palm Recognition Dataset, offer fully cleared collections comprising 2,000 diverse subjects across 200 mobile device models[31].

Several open-source repositories provide implementations of these biometric algorithms. The CompNet architecture (github.com/xuliangcs/compnet) implements learnable Gabor filters for palmprint verification[21]. DeepMPV+ (github.com/ruofei7/Palmprint_Recognition) provides a lightweight MobileFaceNet backbone adapted with ArcFace for mobile devices[12]. The RobustPalmRoi project (github.com/leosocy/RobustPalmRoi) provides pre-trained models for detecting palm bounding boxes and extracting ROIs[34].

For commercial initiatives launching without proprietary training imagery, these legal constraints highlight two deployment strategies:

Deploying Competitive Code written in native C++ or Python (using OpenCV or NumPy) provides a legally clean path forward[13]. Because CompCode relies on deterministic Gabor mathematics without learned weights, it avoids commercial dataset restrictions entirely[16].

Alternatively, teams using deep metric learning must verify that pre-trained model backbones rely on commercially cleared datasets (such as AxonData) or general visual features from ImageNet, and then fine-tune them as operational enrollment databases grow[28].

Presentation Attack Detection and RGB Liveness Verification

Touchless palm systems using standard RGB cameras are vulnerable to Presentation Attacks

(spoofing), including printed photographs, curved paper printouts, and digital displays[35]. Without dedicated near-infrared (NIR) or time-of-flight (ToF) depth sensors, presentation attack detection (PAD) must rely on monocular spatial, temporal, and physiological cues[8].



3D Landmark Planarity and Curvature Analysis

MediaPipe outputs relative depth coordinates (z) alongside 2D coordinates for all 21 hand joints[6]. While single-camera depth estimations contain noise, they reliably capture structural shape[6]. When an attacker presents a flat printed photo or a tablet screen, the coordinates lie on a common plane[38].

A live hand exhibits natural anatomical curvature; the palmar arch cups inward, and the thenar and hypothenar muscles elevate the edges[38]. The system fits a least-squares regression plane across key structural landmarks ($P_0, P_1, P_5, P_9, P_{13}, P_{17}$):

$$Ax + By + Cz + D = 0$$

The root-mean-square error (RMSE) of the orthogonal distances from each landmark to the plane measures spatial variation[38]:

$$\sigma_{\text{planarity}} = \sqrt{\frac{1}{K} \sum_{i \in \mathcal{K}} \frac{(Ax_i + By_i + Cz_i + D)^2}{A^2 + B^2 + C^2}}$$

If the variance falls below a calibrated limit ($\sigma_{\text{planarity}} < \epsilon_{\text{flat}}$), the input is rejected as a flat 2D spoof[38].

Remote Photoplethysmography (rPPG) Pulse Extraction

Live vascularized tissue exhibits periodic color variations driven by blood volume pulse (BVP) cycles during cardiac contractions[8]. Over a two-second capture window (30 to 60 frames at 30 FPS), the MediaPipe ROI remains locked onto the palmar skin[5]. The system calculates the mean intensity of the green spectrum $\bar{g}(t)$, where hemoglobin exhibits peak optical absorption[8]:

$$\bar{g}(t) = \frac{1}{|\text{ROI}|} \sum_{(x,y) \in \text{ROI}} I_G(x, y, t)$$

The raw signal is detrended and passed through a bandpass Butterworth filter with cutoffs at $[0.75, 3.0]$ Hz, isolating normal cardiac frequencies between 45 and 180 beats per minute[8]. Applying a Fast Fourier Transform (FFT) reveals the spectral power density[8]. A genuine hand produces a distinct cardiovascular frequency peak:

$$\text{SNR}_{\text{pulse}} = \frac{\int_{f_0-\Delta}^{f_0+\Delta} P(f)df}{\int_{0.75}^{3.0} P(f)df - \int_{f_0-\Delta}^{f_0+\Delta} P(f)df} > \tau_{\text{pulse}}$$

Static photographs, printed paper artifacts, and passive screen replays lack this cardiac frequency peak and are discarded before reaching feature extraction[8].

Interactive Behavioral Challenges

To protect against recorded video replay attacks, the verification pipeline can require active, randomized behavioral challenges[9]. Users are prompted to perform simple micro-gestures, such as spreading their fingers or rotating their hand along the vertical axis by $\pm 10^{\circ}$ [9]. MediaPipe tracks the angular movement across intermediate joints (such as the vector between P_5 and P_8 relative to P_9 and P_{12}) in real time[8]. Once the challenge movement is validated within a pseudo-random time window (e.g., 1.5 seconds), the system captures the canonical palm frame, crops the ROI, and completes verification[7].

End-to-End Implementation Pipeline

The complete biometric verification system links MediaPipe tracking directly to identity verification, operating across distinct processing stages:





The pipeline begins with camera capture, ingesting RGB video frames at 1280×720 resolution at 30 FPS[5]. Each frame passes to the MediaPipe Hand Landmarker task, which uses BlazePalm to localize the hand and regresses 21 3D landmarks with associated detection and tracking confidence metrics[6]. Frames falling below a 0.80 tracking confidence threshold or displaying significant motion blur (indicated by a Laplacian variance $\sigma^2_{\text{Laplacian}} < 100$) are filtered out[7].

Frames that pass initial filtering undergo quality and pose validation. Scale gating requires the palm width span $\|P_5 - P_{17}\|_2$ to occupy between 25% and 50% of the horizontal frame width, confirming appropriate hand distance from the lens without perspective distortion[5]. Pose validation ensures the calculated orientation angle θ remains within $\pm 45^\circ$ of vertical[9]. Handedness classification confirms the palmar surface faces the sensor rather than the dorsal side[5].

Once validated, the frame undergoes dynamic ROI extraction. The image is rotated around knuckle landmark P_9 by $-\theta$ to bring the hand into a vertical orientation[9]. The palm width L_{palm} determines both the center point C_{roi} and the crop dimension S_{roi}, producing a square image that is resized to 128×128 pixels and processed with CLAHE normalization[7].

The extracted ROI proceeds through presentation attack detection. The system fits a least-squares plane to landmarks $P_0, P_1, P_5, P_9, P_{13}, P_{17}$ and rejects targets with a planarity variance $\sigma_{\text{planarity}} < \epsilon_{\text{flat}}$ [11]. Concurrently, green-channel values across a rolling

30-frame window are filtered within the cardiac frequency range ($[0.75, 3.0]$ Hz) to detect micro-vascular pulses[8]. Inputs failing either test are flagged as presentation attacks and dropped[37].

Frames that clear anti-spoofing checks advance to feature extraction. Under the zero-data strategy, the normalized 128×128 ROI is convolved with the six-orientation Gabor filter bank, generating three binary bit-planes[17]. Under the deep metric strategy, the ROI passes through a pre-trained backbone (such as CompNet or MobileFaceNet), which outputs an L_2 -normalized 512-dimensional vector[12].

In the final verification stage, the input template is compared against the stored enrollment record. Competitive Code templates are matched using translation-compensated bitwise Hamming distance across ± 2 pixels, granting verification if $D_{\text{Hamming}} \leq 0.35$ [13]. Deep feature embeddings are matched using Cosine Similarity, granting verification if $S \geq 0.65$ [12]. For one-to-many (1:N) identification, parallel matrix multiplications or bit-parallel XOR sweeps identify matching enrolled profiles within low millisecond latencies[13].

Comparative Benchmark Performance Across Verification Pipelines

Evaluating verification pipelines across standard contactless palm databases highlights the trade-offs between deterministic coding schemes and deep metric networks[13]:

Algorithm Architecture	Core Feature Representation	Benchmark Database	Correct Recognition Rate (CRR)	Equal Error Rate (EER)	Match Latency	Training Data Requirement
Competitive Code (CompCode)	6-Directional Gabor Response[18]	Tongji Contactless[13]	98.78% [13]	$\approx 0.12\%$	12.5 ms[13]	**Zero (Deterministic)**[16]
Competitive Code (CompCode)	6-Directional Gabor Response[18]	IIT Delhi (IITD)[24]	98.91% [30]	$\approx 0.40\%$	11.8 ms	**Zero (Deterministic)**[16]
CompNet	Learnable Gabor ConvNet[2]	Tongji Contactless[42]	99.72% [24]	0.075% [21]	18.0 ms	Pre-trained[21]

	[1]					
CompNet	Learnable Gabor ConvNet[21]	CASIA Contactless[24]	99.67% [24]	0.082% [24]	17.5 ms	Pre-trained[21]
DeepMPV+ (ArcFace)	MobileFaceNet (512-D)[12]	MPD Smartphone[12]	99.45% [12]	0.088% [12]	15.2 ms[12]	Pre-trained[12]
HDC-Net (SupCon)	Heterogeneous Dual-Stream[22]	Tongji Contactless[22]	99.85%	0.140% [22]	22.0 ms	Pre-trained[22]
PalmMobile SDK (COTS)	Commercial Deep Embeddings[15]	Tongji Contactless[15]	100.0% [15]	$< 0.010\%$ [15]	20.0 ms	Proprietary[15]

Unsupervised approaches, such as Competitive Code, achieve strong recognition rates ($> 98.7\%$) without training data[13]. This allows teams to deploy an effective matching pipeline without the intellectual property risks associated with academic benchmark models[16].
Deep metric models, such as CompNet and ArcFace, achieve lower Equal Error Rates ($0.04\% - 0.08\%$) and show higher resilience to sub-pixel misalignment[12]. However, these architectures require careful validation of dataset licensing and model lineage before commercial release[3].

Engineering Conclusions and Staged Implementation Plan

Building a commercial, contactless palm biometric verification system without initial training images is achievable by combining open-source landmark frameworks with classical feature extraction:
MediaPipe Hands should be deployed under its Apache 2.0 license to handle real-time hand acquisition, pose gating, and dynamic ROI segmentation[1]. This avoids the need to train complex object detection models from scratch[6].
For the initial commercial deployment, feature matching can be implemented using Competitive Code (CompCode)[13]. Because CompCode relies on deterministic 2D Gabor equations rather than learned weights, it eliminates the need for proprietary training data and sidesteps the

restrictive licenses of academic benchmarks[13].
To protect against presentation attacks using standard RGB cameras, systems should combine 3D landmark planarity checks (to detect flat printouts or screens) with temporal rPPG green-channel pulse detection[8].
As users enroll and the system accumulates proprietary images under appropriate privacy frameworks, teams can transition to fine-tuning lightweight neural architectures (such as MobileFaceNet or CompNet) with ArcFace loss[12]. This migration path achieves high biometric accuracy ($EER < 0.08\%$) while maintaining complete commercial and intellectual property compliance throughout the product lifecycle[3].

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